WORKED SOLUTION

STEP III 2000, Question 2

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q2

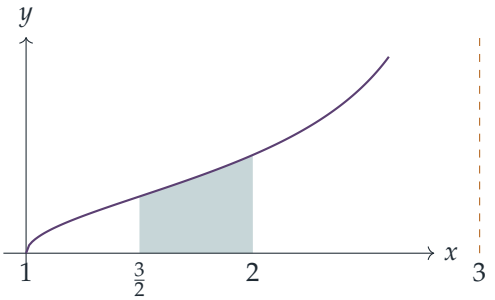
The key idea

A square root of a quotient like $(x-1)/(3-x)$ is begging for the half-angle formulae: with $x = 2 - \cos\theta$ the top becomes $1 - \cos\theta$ and the bottom $1 + \cos\theta$, and their ratio is a perfect square, $\tan^2 \frac{\theta}{2}$. For the general result we don't need to redo the work. A linear change of variable carries the interval from a to b onto the interval from 1 to 3, and turns the general integral into a multiple of the first one.

Solution

The first integral

The integrand is defined for $1 \leq x < 3$, and our range $\frac{3}{2} \leq x \leq 2$ lies inside it. The shaded area below is the integral we want.



Put $x = 2 - \cos\theta$, so $\mathrm{d}x = \sin\theta\,\mathrm{d}\theta$. When $x = \frac{3}{2}$, $\cos\theta = \frac{1}{2}$, and when $x = 2$, $\cos\theta = 0$, so we take θ from $\frac{\pi}{3}$ to $\frac{\pi}{2}$. Then

$$x - 1 = 1 - \cos\theta = 2\sin^2\tfrac{\theta}{2}, \qquad 3 - x = 1 + \cos\theta = 2\cos^2\tfrac{\theta}{2},$$

so

$$\left(\frac{x-1}{3-x}\right)^{\frac{1}{2}} = \left|\tan\tfrac{\theta}{2}\right| = \tan\tfrac{\theta}{2},$$

because $0 < \frac{\theta}{2} < \frac{\pi}{2}$ on our range, so the tangent is positive.

Ask yourself Once the integrand is $\tan\frac{\theta}{2}\sin\theta$, how do we avoid integrating a tangent?

Write $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$. The cosine cancels:

$$\tan\tfrac{\theta}{2}\,\sin\theta = \frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}\cdot 2\sin\tfrac{\theta}{2}\cos\tfrac{\theta}{2} = 2\sin^2\tfrac{\theta}{2} = 1-\cos\theta.$$

Therefore

$$\begin{aligned}\int_{3/2}^{2}\left(\frac{x-1}{3-x}\right)^{\frac{1}{2}}\mathrm{d}x &= \int_{\pi/3}^{\pi/2}(1-\cos\theta)\,\mathrm{d}\theta = \Big[\theta-\sin\theta\Big]_{\pi/3}^{\pi/2}\\ &= \left(\frac{\pi}{2}-1\right)-\left(\frac{\pi}{3}-\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}-1+\frac{\sqrt{3}}{2}\\ &= \frac{\pi+3\sqrt{3}-6}{6}.\end{aligned}$$

The general integral

The first integral is the case $a=1$, $b=3$, with $p = \frac{3+3}{4} = \frac{3}{2}$ and $q = \frac{1+3}{2} = 2$. So we look for a linear substitution that sends $u=1$ to $x=a$ and $u=3$ to $x=b$:

$$x = a + \tfrac{1}{2}(b-a)(u-1), \qquad \mathrm{d}x = \tfrac{1}{2}(b-a)\,\mathrm{d}u.$$

Then

$$x-a = \tfrac{1}{2}(b-a)(u-1), \qquad b-x = (b-a)-\tfrac{1}{2}(b-a)(u-1) = \tfrac{1}{2}(b-a)(3-u),$$

and since $b-a \neq 0$ the common factor cancels:

$$\frac{x-a}{b-x} = \frac{u-1}{3-u}.$$

For the limits: $u=\frac{3}{2}$ gives $x = a+\frac{1}{4}(b-a) = \frac{3a+b}{4} = p$, and $u=2$ gives $x = a+\frac{1}{2}(b-a) = \frac{a+b}{2} = q$. Since $b-a>0$, the substitution is increasing, so the limits keep their order. Hence

$$\begin{aligned}\int_p^q\left(\frac{x-a}{b-x}\right)^{\frac{1}{2}}\mathrm{d}x &= \frac{b-a}{2}\int_{3/2}^{2}\left(\frac{u-1}{3-u}\right)^{\frac{1}{2}}\mathrm{d}u\\ &= \frac{b-a}{2}\cdot\frac{\pi+3\sqrt{3}-6}{6} = \frac{(b-a)(\pi+3\sqrt{3}-6)}{12},\end{aligned}$$

using the first result. This is what we had to show.

(Equivalently, you could substitute $x = \frac{a+b}{2} - \frac{b-a}{2}\cos\theta$ directly. The limits are again $\frac{\pi}{3}$ and $\frac{\pi}{2}$, and the working is the same with an extra factor $\frac{b-a}{2}$.)

Checks

- Numerically, $\frac{1}{6}(\pi + 3\sqrt{3} - 6) \approx 0.38962$. Simpson's rule applied to the original integrand on $[\frac{3}{2}, 2]$ gives the same value to ten decimal places.
- Plausibility: the integrand rises from $\sqrt{1/3} \approx 0.577$ to 1 over an interval of length $\frac{1}{2}$, so the area must lie between about 0.29 and 0.5. It does.
- The general formula agreed with Simpson's rule for 20 random choices of $a < b$, including negative a.
- Scaling: stretching the interval $[a, b]$ by a factor k leaves the integrand's values unchanged but stretches $[p, q]$ by k. So the answer must be proportional to $b - a$, as it is.

Common slips

- Forgetting $\mathrm{d}x = \sin\theta\,\mathrm{d}\theta$ and trying to integrate $\tan\frac{\theta}{2}$ alone with respect to θ.
- Taking the limits as $\theta = -\frac{\pi}{3}$ or $\frac{5\pi}{3}$. Any choice must make θ increase from $\frac{\pi}{3}$ to $\frac{\pi}{2}$ through a range where $\sin\theta > 0$ and $\tan\frac{\theta}{2} > 0$.
- Missing the factor $\frac{1}{2}(b - a)$ from $\mathrm{d}x$ in the general case, which gives 6 in the denominator instead of 12.

Beyond STEP

The second part shows that a definite integral transforms predictably under an affine change of variable. This is why quantities like this one depend on a and b only through the length $b - a$. The integrand itself is a beta-function-type integrand: over the whole interval, $\int_a^b \left(\frac{x-a}{b-x}\right)^{1/2}\mathrm{d}x = \frac{\pi}{2}(b - a)$, and the functions $\int_0^1 t^{\alpha-1}(1-t)^{\beta-1}\,\mathrm{d}t$ recur throughout probability and analysis. Blitzstein and Hwang, *Introduction to Probability*, meets them as the normalising constants of the Beta distribution. Bartle and Sherbert, *Introduction to Real Analysis*, proves the substitution rule carefully.

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